Physics P2: Motion, Forces, and Conservation

MAIS Physics & Astronomy Club (PAC)

Newton’s second law **F** = **ma** determines the entire trajectory of a mechanical system in principle, yet integrating forces instant by instant is often the hardest route. Conservation of energy and conservation of momentum extract global information—relating initial and final states directly—even when intermediate forces are complicated or rapidly varying.

1. Kinematics and Reference Frames

Position **r**(**t**), velocity **v** = **dr**/**dt**, and acceleration **a** = **dv**/**dt** are vectors. Under constant gravitational acceleration **g**, horizontal and vertical motions decouple, while switching to another inertial frame moving at constant velocity **u** shifts all velocities by $\mathbf{v}' = \mathbf{v} - \mathbf{u}$ without altering accelerations or forces.

Example 1. Projectile A is launched from ground level toward an elevated target, and at the exact same instant a ball B at the target is dropped from rest. Assuming no air resistance and that A’s initial velocity vector points directly at B’s starting position, do they collide?

Solution. Switch to the free-falling frame accelerating downward at **g**. In this frame gravity disappears: ball B floats motionless at its starting point, while projectile A travels in a straight line at constant speed $\mathbf{v}_0$ aimed directly at B. Provided A is fast enough to reach B before hitting the ground, they always collide—regardless of the launch angle!

[1] Problem 1. Three identical stones are thrown from the edge of a cliff of height **h** with the same initial speed $\mathbf{v}_0$: stone 1 at angle $+\alpha$ above the horizontal, stone 2 horizontally, and stone 3 at angle $-\alpha$ below the horizontal. Neglect air resistance.

(a) Compare the speeds $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ with which the three stones strike the ground below.

(b) Find the difference $\mathbf{t}_1 - \mathbf{t}_3$ between the total flight times of stones 1 and 3, and explain physically why your result is independent of the cliff height **h**.

2. Forces and Circular Constraints

Whenever an object follows a curved path of local radius **R** at speed **v**, its acceleration has a radial (centripetal) component $\mathbf{v}^2/\mathbf{R}$ directed toward the center of curvature and a tangential component **dv**/**dt** along the path:

$$\mathbf{a_r} = (\mathbf{v}^2)/\,(\mathbf{R}),\ \mathbf{a_t} = (\mathbf{dv})/\,(\mathbf{dt}). \tag{1}$$

Remark. “Centripetal force” is never a separate arrow on a free-body diagram! Draw only real physical interactions—gravity, normal contact forces, tension, friction—and set their net inward component equal to $\mathbf{mv}^2/\mathbf{R}$.

[2] Problem 2. A small bead of mass **m** slides from rest down a frictionless ramp from height **h** above the ground and enters a vertical circular loop-the-loop of radius **R** whose bottom touches the ground.

(a) Find the minimum release height $\mathbf{h}_{\text{min}}$ (in terms of **R**) so that the bead stays in contact with the track at the very top of the loop.

(b) When released from $\mathbf{h} = \mathbf{h}_{\text{min}}$, find the magnitude and direction of the bead’s **total** acceleration vector when it reaches the horizontal midpoint of the loop (at height **R** above the ground).

3. Conservation of Energy and Momentum

For an isolated system with no external net force, total momentum $\mathbf{P} = \sum \mathbf{m_i v_i}$ is conserved—even during violent inelastic collisions where mechanical energy is converted into heat. When all forces are conservative (elastic collisions, gravity, ideal springs), total mechanical energy $\mathbf{E} = \mathbf{K} + \mathbf{U}$ is conserved as well.

$$\mathbf{K} = (1)/(2)\,\mathbf{mv}^2 = (\mathbf{p}^2)/(2\mathbf{m}),\ \Delta\mathbf{E}_{\text{mech}} = \mathbf{W}_{\text{non-cons}}. \tag{2}$$

[2] Problem 3 (The gravitational slingshot in miniature). Hold a light tennis ball of mass **m** just above a heavy basketball of mass $\mathbf{M} \gg \mathbf{m}$, and drop both together from rest at height **h** above a hard floor. Assume all collisions are perfectly elastic and vertical.

(a) Just after the basketball rebounds off the floor at upward speed $\mathbf{v}_0 = \sqrt{(2\mathbf{gh})}$, switch into the basketball’s rest frame to find the rebound speed of the tennis ball in the room frame (in the limit $\mathbf{M} \gg \mathbf{m}$).

(b) To what maximum height (in terms of **h**) does the tennis ball rise? Explain briefly how space probes use the exact same mechanism when flying past Jupiter.

[3] Problem 4 (Falling chain on a scale). A uniform flexible chain of total mass **M** and length **L** is held vertically so that its bottom tip barely touches the pan of an electronic force scale. At $\mathbf{t} = 0$ the chain is released from rest; each link comes to rest immediately upon hitting the pan without bouncing.

(a) At an instant when a length $\mathbf{y} < \mathbf{L}$ of the chain has already fallen onto the pan, find the speed **v** of the falling upper portion and the rate **dm**/**dt** at which mass strikes the pan.

(b) Show that the upward force exerted by the scale is $\mathbf{F}(\mathbf{y}) = (3\mathbf{Mg})/(\mathbf{L})\,\mathbf{y}$ for $0 < \mathbf{y} < \mathbf{L}$. What is the maximum reading right before the top link lands, and what does the reading drop to afterward?